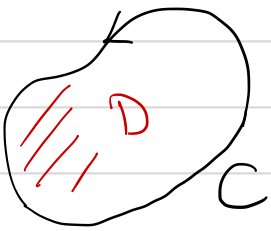
Lecture 16

Green's theorem (16.4)

$$\boxed{\int_C P\,dx + Q\,dy = \iint_D Q_x - P_y \, dA}$$



C: closed curve (positively oriented)
D: region bounded by C.

Area from Green's theorem

Choose P, Q so $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$.

Then:

$$\boxed{\int_C P\,dx + Q\,dy = \underbrace{\iint_D 1 \, dA}_{\text{Area of } D.}}$$

Examples: $(P,Q) = (0, x)$

$(P,Q) = (-y, 0)$

$(P,Q) = \frac{1}{2}(-y, x)$

$\vdots$

Example: Ellipse $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 \leq 1$

$$\text{Area} = \frac{1}{2}\int_C -y\,dx + x\,dy$$

$x = a\cos t \qquad y = b\sin t$

$0 \leq t \leq 2\pi$

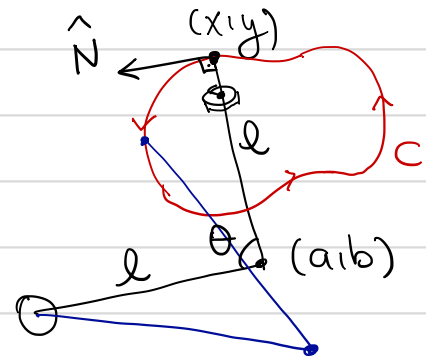
Green's theorem

$$\text{Area} = \frac{1}{2}\int_0^{2\pi} \left[-b\sin t \cdot (-a\sin t) + a\cos t \cdot (b\cos t)\right] dt$$

$$= \frac{ab}{2}\int_0^{2\pi} \sin^2 t + \cos^2 t \, dt$$

$$= ab\pi \;//$$

Planimeter



$\frac{1}{\ell}$(Area inside C) =

= distance (x,y) moves in the direction perpendicular to the arm.

(measured by wheel)

$$\int_C \hat{N} \cdot d\vec{r}$$

Why: ① Green's theorem

② $\hat{N} = \langle P, Q \rangle$

$$\Rightarrow \boxed{Q_x + P_y = 1/r}$$

$$\int_C \underbrace{\hat{N}}_{\langle P,Q\rangle} \cdot \underbrace{d\vec{r}}_{\langle dx,dy\rangle} = \int_C P\,dx + Q\,dy$$

$$= \iint_D \underbrace{Q_x - P_y}_{1/\ell}\, dA$$

$$= \frac{1}{\ell}\,\text{Area}(D)$$

<u>How to check</u> $Q_x - P_y = 1/\ell$:

$\hat{N} = \frac{1}{\ell}(-(y-b),\, x-a) = (P, Q)$

↳ Normal to arm, length 1

(Because: $(x-a,\, y-b) \parallel$ arm)

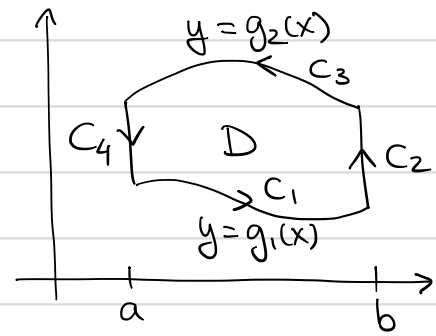
Here $a = a(x,y)$, $b = b(x,y)$ satisfy:

$$\begin{cases} (x-a)^2 + (y-b)^2 = \ell^2 & (*) \\ a^2 + b^2 = \ell^2 & (**) \end{cases}$$

$$Q_x = \frac{1}{\ell}(1 - a_x)$$
$$P_y = -\frac{1}{\ell}(1 - b_y)$$

Check $Q_x - P_y = 1/\ell$ by taking partials of both sides of $(*)$ and $(**)$.

<u>Proof of Green's theorem when D is both type I & II</u>



Type I

To check:

$$\int_C P\,dx = -\iint_D \frac{\partial P}{\partial y}\, dA \qquad (*)$$

$$\int_C Q\,dy = \iint_D \frac{\partial Q}{\partial x}\, dA \qquad (**)$$

<u>RHS</u> $(*)$:

$$\iint_D -\frac{\partial P}{\partial y}\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} -\frac{\partial P}{\partial y}\, dy\, dx$$

$$= \int_a^b -P(x, g_2(x)) + P(x, g_1(x))\, dx$$

(C_3 goes from right to left)

$$= +\int_{C_3} P\,dx + \int_{C_1} P\,dx +$$

$$\underbrace{+\int_{C_2} P\,dx + \int_{C_4} P\,dx}_{=0 \text{ because } C_2 \,\&\, C_4 \text{ are vertical.}}$$

$$= \int_C P\,dx = \text{LHS } (*) \qquad \blacksquare$$

$\hat{N} = \frac{1}{\ell}(-(y-b), x-a) = (P, Q)$

↳ Normal to arm, length 1

(Because: $(x-a, y-b) \parallel$ arm)

To know

$$(a,b) = (a(x,y), b(x,y))$$

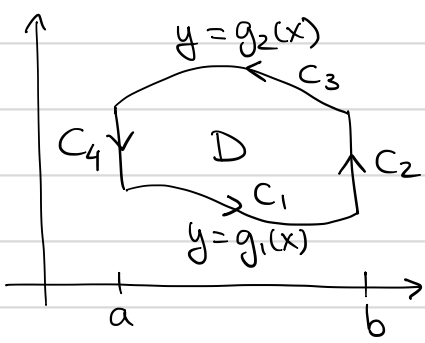
Solve

$$\begin{cases} (x-a)^2 + (y-b)^2 = \ell^2 \\ a^2 + b^2 = \ell^2 \end{cases}$$

$(0 \leq \theta \leq \pi)$

Then: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (\dots) = \frac{1}{\ell}$ //

Proof of Green's theorem when D is both type I & II



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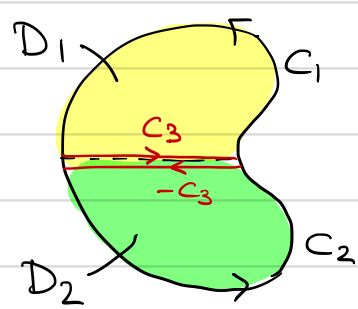
More general regions



Green's for D_1:

$$\int_{C_1 \cup C_3} P\,dx + Q\,dy = \iint_{D_1} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Green's for D_2:

$$\int_{C_2 \cup -C_3} P\,dx + Q\,dy = \iint_{D_2} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

$\Rightarrow$ Green's for $D_1 \cup D_2$.

$$\int_{C_1 \cup C_2} P\,dx + Q\,dy = \iint_{D_1 \cup D_2} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

(C_3 and $-C_3$ cancel)

[Check (*) by breaking D into regions of type I, and check (**) by breaking D into regions of type II.]